

Federal Spending Analysis

Where Do U.S. Federal Award Dollars Go?

An analysis of federal contracts, grants, agencies, and geographic distribution

Kristoffer Fields

Georgia Institute of Technology

Master of Science in Analytics

September 2026

Project Overview

The federal government distributes trillions of dollars each year through programs including contracts, grants, direct payments, and other forms of federal assistance. Understanding where these funds go provides insight into federal policy priorities and the role government spending plays across agencies, industries, organizations, and geographic regions. The purpose of this project is to examine available public spending data from USAspending.gov to develop further insight on the distribution of federal award dollars.

Research Question

How does the federal government distribute contract and grant funding across agencies and states, and what patterns emerge in how that funding is allocated?

Objectives

- Identify which federal agencies account for the largest amounts of contract and grant spending.
- Compare the distribution of federal contracts and grants.
- Examine differences in federal award spending across U.S. states.
- Use quantitative analysis and visualization to communicate federal spending patterns clearly.

Data & Tools

Data Source: USAspending.gov

Analysis: Python, pandas

API Access: requests

Visualization: Matplotlib, Plotly

Understanding Federal Spending

It is important to understand how federal funding is distributed from decisions in congress to the organization that ultimately receives the funding. Essentially, congress first has to approve the funding, then federal agencies manage that funding, agencies issue the grants, and finally recipients receive the funding.

Appropriations

Federal spending appropriations are laws for funding operations. They are the legal authority that Congress uses to disburse federal funding for specific purposes.

Each year, the president submits a proposed federal budget to Congress outlining the administration's funding priorities for the upcoming fiscal year. Congress reviews these proposals and develops appropriations bills that determine how much funding federal agencies and programs will receive. These bills must be passed by Congress and signed into law before the funds can generally be used.

Furthermore, the year that congress uses is the Fiscal year which is from October 1st to September 30th. This is especially important for this project because it is how my data will be organized.

Contracts and Grants

A federal contract is when the government is buying a good or service from a third-party to carry out its own government functions. For example, the government may create a contract with a technology company in order to provide cybersecurity services. On the contrary, a federal grant is when the government is using the money to support a greater public good such as university research or infrastructure.

Simply, a contract is when the government purchases something and a grant is when it funds an activity or public purpose. This distinction is important for this project. I will examine how federal dollars are distributed under both types of awards. Additionally, I will examine which recipients receive the most money, where does the money go geographically, and which agencies spend the most money.

Recipients and Federal Awards

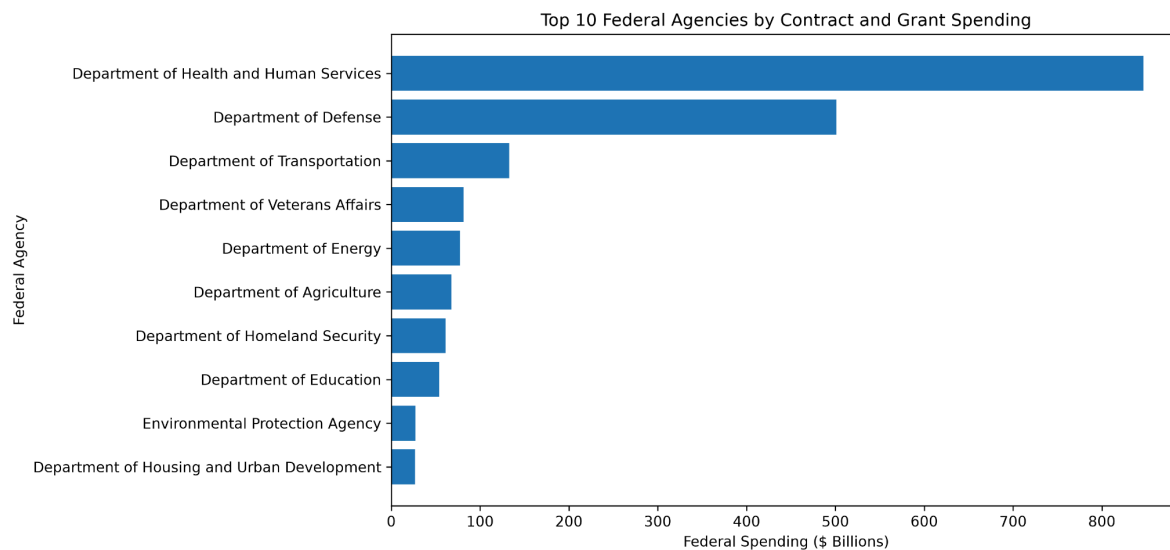
Federal award recipients are organizations or entities that receive funding from the federal government through contracts, grants, or other financial assistance. These may include private companies, universities, nonprofit organizations, and state or local governments. A recipient may receive multiple awards from different federal agencies, depending on the work or program funded. Understanding who can receive federal awards helps provide context for how contracts and grants move from federal agencies into the broader economy. While this project focuses mainly on agencies, award types, geography, and spending trends, recipients are an important part of the overall federal funding process

Data Source and Methodology

The data used in this project comes from USAspending.gov, the official source for information on federal spending in the United States. USAspending.gov provides information on federal awards, including contracts and grants, as well as the agencies that issue the awards, the recipients that receive them, and the geographic locations where the funding is distributed.

For this analysis, the data will be organized by federal fiscal year and will focus primarily on federal contracts and grants. Python will be used to collect, clean, and analyze the data. The analysis will examine federal spending by agency, award type, recipient, and geographic location in order to identify patterns in how federal funding is distributed.

Figure 1



Source: USAspending.gov, Fiscal Year 2025 data. Retrieved September 2026.

Key Findings

- The top two agencies accounted for 71.8% of total contract and grant spending, showing that federal award spending was highly concentrated.
- The Department of Health and Human Services exceeded the Department of Defense by \$345.6 billion in contract and grant spending.
- Spending dropped by 73.5% from the second-ranked agency to the third-ranked agency, demonstrating the large gap between the top two agencies and the rest.

Analysis

Federal contract and grant spending is heavily concentrated among a small number of agencies. The Department of Health and Human Services had the highest amount in the dataset at about \$847 billion, followed by the Department of Defense at roughly \$501 billion. After those two agencies, spending falls sharply to about \$133 billion for the Department of Transportation.

The size of these gaps shows that federal award spending is not distributed evenly across agencies. In fact, the top two agencies accounted for 71.8% of the total contract and grant spending in the dataset. This makes HHS and DOD clear outliers compared with the rest of the agencies shown.

It is also important to note that this chart represents contract and grant spending only, not each agency's total federal budget. The chart is therefore best understood as showing which agencies play the largest role in distributing these types of federal awards.

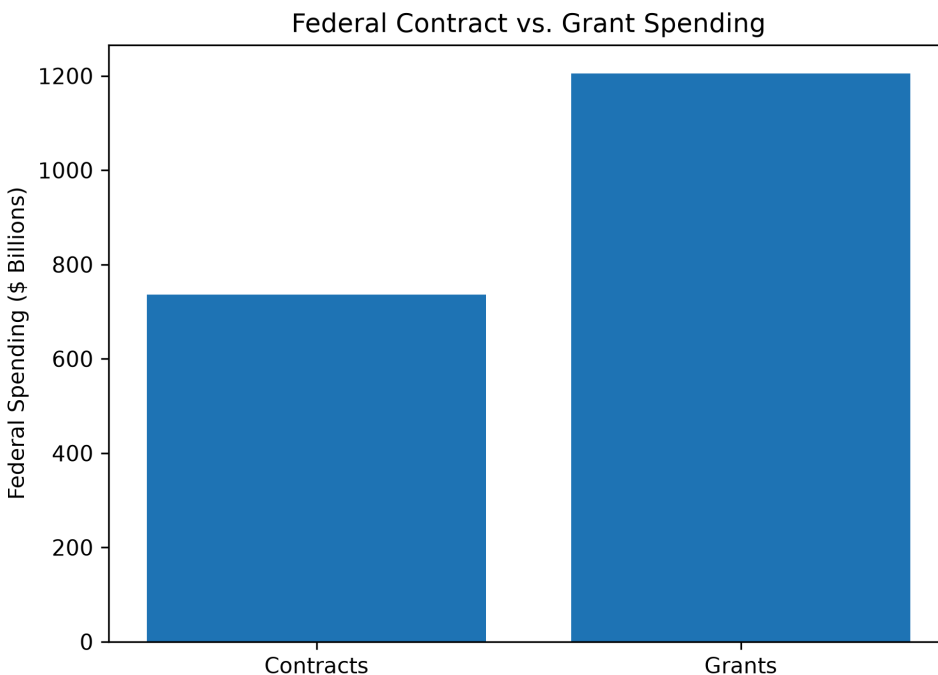
Why are HHS and DoD so much larger?

The concentration of spending among agencies such as the Department of Health and Human Services and the Department of Defense reflects the scale of their missions and programs. HHS administers large health and social-service programs and distributes substantial funding through grants to states, universities, healthcare organizations, and other recipients. The Department of Defense, meanwhile, relies heavily on contracts to purchase military equipment, technology, research, logistics, and other services. Because these agencies oversee some of the federal government's largest programs and procurement activities, their award spending is substantially higher than that of most other agencies.

A big reason for this concentration is that HHS and DOD both manage extremely large federal responsibilities, but in different ways. HHS distributes large amounts of funding for healthcare, research, and assistance programs, while DOD spends heavily on equipment, technology, construction, and services. That helps explain why these two agencies stand so far above the rest. ¹

¹ U.S. Department of Health and Human Services. "Grants and Contracts." Accessed September 2026.

Figure 2



Source: USAspending.gov, Fiscal Year 2025 data. Retrieved September 2026.

Analysis

The federal government spent a lot more through grants than through contracts. Grant spending was about \$1.2 trillion, while contract spending was around \$740 billion. That is a difference of roughly \$460 billion, meaning grant spending was about 60% higher.

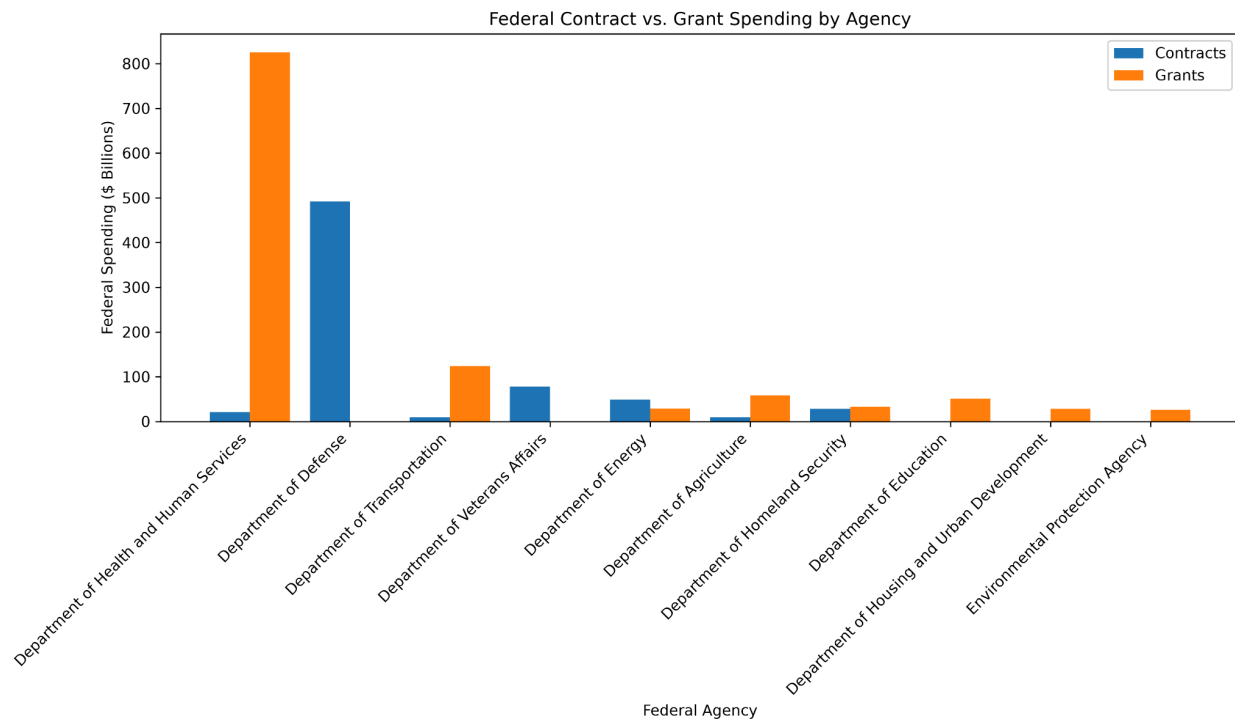
This difference comes from the fact that contracts and grants serve different purposes. Contracts are generally used when the government buys goods or services, while grants support programs and activities such as research, infrastructure, healthcare, and other forms of public assistance.

At the same time, the overall totals do not tell us which agencies are driving that difference. Some agencies may rely heavily on grants, while others may depend much more on contracts. Looking at spending by agency helps show those differences more clearly and gives a better picture of how federal funding is actually distributed.

Why is grant spending higher than contract spending?

Grant spending is higher because grants are used to fund broad public purposes rather than simply purchase goods and services for the federal government. Grants can support state and local governments, universities, nonprofit organizations, healthcare systems, research programs, infrastructure, and other federally authorized activities. Contracts, by comparison, are primarily used when agencies purchase specific goods or services for government use. Because grants can fund large programs implemented across many outside organizations and governments, total grant spending can exceed contract spending.

Figure 3



Source: USAspending.gov, Fiscal Year 2025 data. Retrieved September 2026.

Key Findings

- The Department of Health and Human Services is dominated by grant spending, while the Department of Defense is dominated by contract spending.
- The Department of Transportation also relies heavily on grants, whereas Veterans Affairs and Energy appear more contract-oriented.
- Federal agencies use contracts and grants very differently depending on their missions, meaning the overall federal totals hide major differences in how individual agencies distribute funding.

Analysis

The chart shows that federal agencies use contracts and grants very differently. The Department of Health and Human Services relied mostly on grants, while the Department of Defense relied much more heavily on contracts. This makes sense because HHS often funds health and social programs, while DOD is more likely to purchase equipment, technology, and other services.

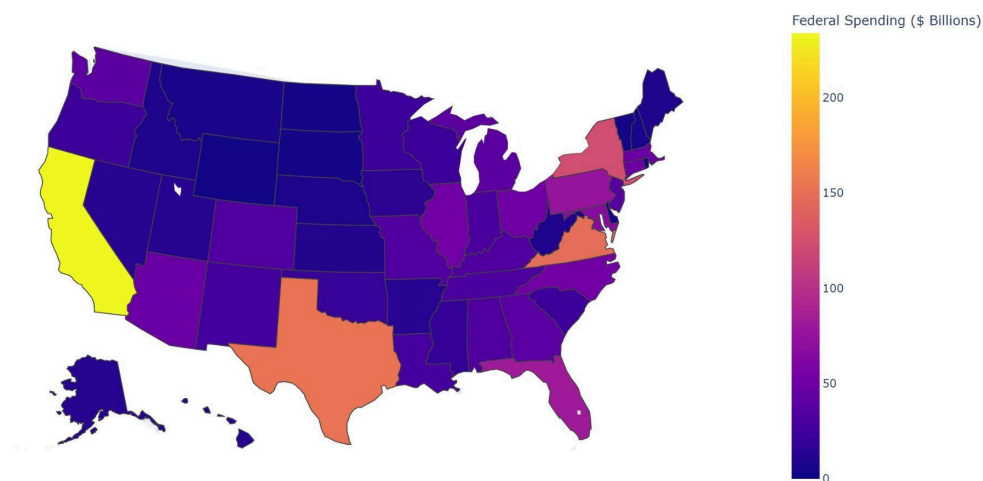
Overall, the chart shows that federal agencies do not use a single standard way to distribute money. The balance between contracts and grants depends largely on what each agency does. It also helps explain why grant spending is higher overall, even though some major agencies, especially the Department of Defense, are much more contract-focused.

Why do agencies use contracts and grants differently?

The differences across agencies largely reflect their missions. Agencies responsible for purchasing equipment, technology, construction, research services, or operational support tend to rely more heavily on contracts. Agencies that primarily fund public programs, research institutions, state and local governments, or nonprofit organizations are more likely to rely on grants. In other words, the award mechanism follows the type of work the agency is trying to accomplish: contracts purchase something for the government, while grants provide financial assistance to carry out a broader public purpose.²

Figure 4

Federal Contract and Grant Spending by State



Source: USAspending.gov, Fiscal year 2025 data. Retrieved September 2026.

Analysis

Federal contract and grant spending is not evenly distributed across the United States. Some states receive much larger amounts of federal award funding than others, which can reflect factors such as population size, the presence of major federal contractors, research institutions, military facilities, and large state or local government programs. The map helps show that federal spending is geographically concentrated rather than spread uniformly across the country.

One important limitation is that this map shows total spending by state, so larger states may naturally receive more funding. A useful next step is to compare federal spending per capita, which adjusts for population and can reveal states that receive especially high levels of funding relative to their size.

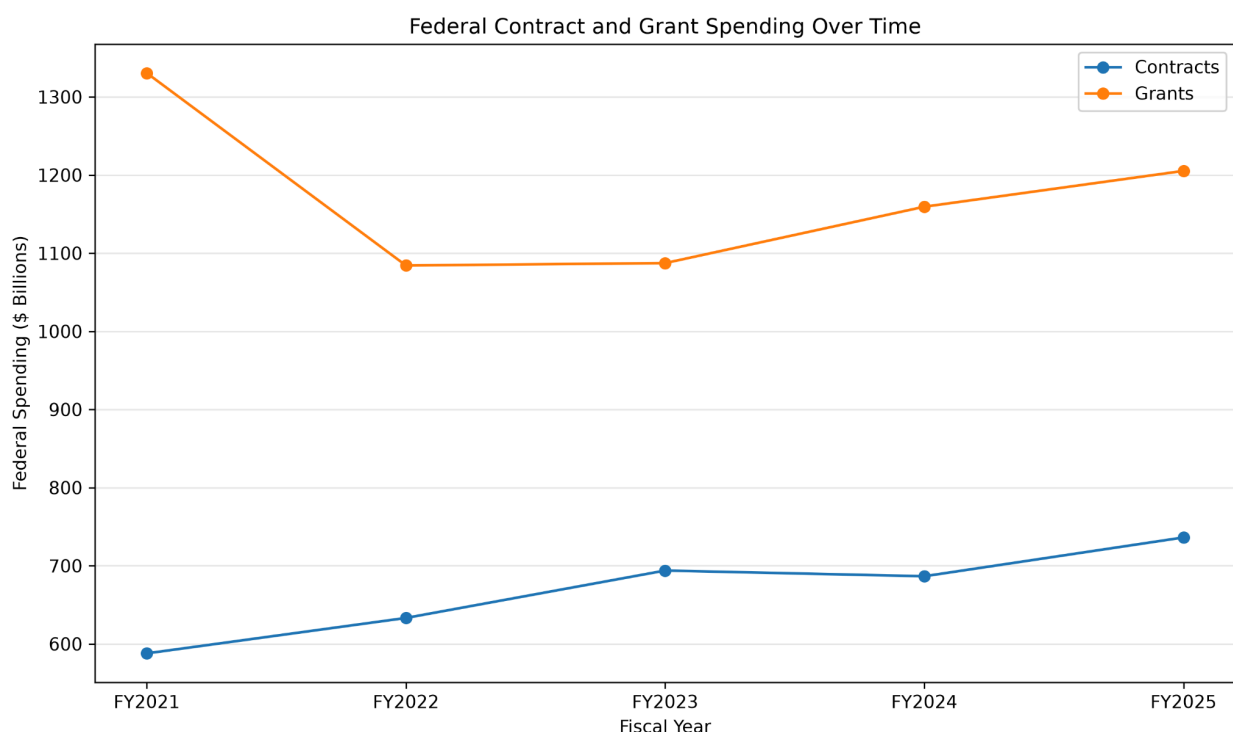
² U.S. Department of Transportation. "Overview of Funding and Financing at USDOT." Accessed September 2026.

Something in particular that was interesting to me is that the midwest and the deep south have less federal contract and grant spending as compared to states like California or the Northeast.

Why is spending concentrated in California and the Northeast?

Federal spending is higher in California and parts of the Northeast for several reasons. These areas have large populations, major universities and research institutions, healthcare systems, defense and technology companies, military facilities, and large state and local government programs that receive federal funding.³ Since the map shows total spending rather than spending per person, larger states and regions with more economic activity are also more likely to receive higher overall amounts.⁴

Figure 5



Source: USAspending.gov, federal contract and grant award data, fiscal years 2021–2025. Retrieved September 2026.

Analysis

What stands out to me is how much more grant spending moves around from year to year. The drop after 2021 is much sharper than anything we see with contracts, which makes sense given how much temporary pandemic relief was still flowing through the federal government at that point. According to the chart, grant spending was highest in 2021 at around 1.33 trillion, as it declines it plateaued in the years 2022 and

³ National Science Foundation, National Center for Science and Engineering Statistics. “Academic Research and Development.” Accessed September 2026.

⁴ U.S. Census Bureau. “QuickFacts: California.” Accessed September 2026.

2023, and then has steadily inclined to the present date.

Contract spending looks much steadier by comparison and seems to reflect the more consistent way agencies buy goods and services over time. It rose from roughly \$590 billion in 2021 to about \$735 billion in 2025, with only a small decline in 2024. Another interesting point is that the gap between grant and contract spending narrowed over the five-year period. Grants were still much higher overall, but contract spending kept increasing while grant spending never returned to its 2021 peak.

Why did spending change over time?

The biggest change in Figure 5 is the decrease in grant spending after 2021. This lines up with the pandemic. Congress approved trillions of dollars in emergency funding for the public. This money was distributed in a multifaceted approach; much of it went to public health, unemployment support, emergency programs, and education. By 2023, much of that money had been spent. The fact that grant spending later begins to rise again suggests that newer federal programs started replacing some of the pandemic-era spending, rather than grants simply continuing to decline.

What I find interesting is that grant spending did not continue to fall. It began to level off in 2022 and then started to rise. This evidence suggests the increase was driven by a separate group of programs, rather than COVID-19 relief.⁵ At the same time, contract spending moved much more gradually, which fits with the fact that agencies continue buying goods and services every year regardless of temporary emergency programs.

⁵ U.S. Government Accountability Office. *COVID-19: Sustained Federal Action Is Crucial as Pandemic Enters Its Second Year*. GAO-21-387, March 31, 2021.

Conclusion

This project demonstrates that federal contract and grant spending is disbursed and concentrated in many different ways. A small number of agencies account for a large share of total award spending. For instance, Health and Human Services and the Department of Defense account for most of the spending. Grants also make up a larger share of spending, although the balance between grants and contracts varies widely by agency.

The geographic analysis shows that spending is not even across the United States. States with larger populations, such as California and the Northeast, disproportionately receive more money. In contrast, states in the Deep South and Midwest receive less. Furthermore, the five-year trend graph shows changes in grant spending over time and how its changes are more volatile than contract spending.

Overall, the project shows that federal spending patterns are shaped by agency responsibilities, the type of funding being used, geography, and broader policy changes over time.

Limitations

This analysis focuses only on federal contracts and grants, so it does not represent all federal spending. The geographic analysis also uses total spending by state, which means larger states may naturally receive more funding. In addition, the charts show patterns and relationships, but they do not prove that any one factor directly caused higher or lower spending.

Tools Used

Python was used to collect, clean, and analyze the data. The project used requests to access the USAspending API, pandas to organize and calculate the data, Matplotlib for charts, and Plotly for the geographic map.

References

1. U.S. Department of Health and Human Services. “Grants and Contracts.” Accessed September 2026.
<https://www.hhs.gov/grants-contracts/index.html>
2. Acquisition.gov. “Defense Federal Acquisition Regulation Supplement (DFARS).” Accessed September 2026.
<https://www.acquisition.gov/dfars>
3. U.S. Department of Transportation. “Overview of Funding and Financing at USDOT.” Accessed September 2026.
<https://www.transportation.gov/grants/dot-navigator/overview-funding-and-financing-usdot>
4. U.S. Census Bureau. “QuickFacts: California.” Population Estimates. Accessed September 2026.
<https://www.census.gov/quickfacts/fact/table/CA/PST045225>
5. U.S. Government Accountability Office. *COVID-19: Sustained Federal Action Is Crucial as Pandemic Enters Its Second Year*. GAO-21-387, March 31, 2021.
<https://www.gao.gov/products/gao-21-387>